

Independent and Correlated Nested Queries

Aim:

To implement and understand Nested Queries in SQL, including Independent and Correlated, with practical examples in a university database scenario.

Procedure

1. Create required tables eg. Students, Departments, Courses, Enrollment.
2. Insert Sample data
3. Write and execute Independent nested queries.
4. Write and execute Correlated nested queries.

Explanation.

A nested query (Subquery) is written inside another query. It is used when the output of one query is required for another query.

Types of nested queries.

1. Independent (Simple) Nested Queries.
 - the inner query runs independently of the outer query.
 - the result of the inner query is passed to the outer query.

Syntax:

```
SELECT Column-List
FROM table
WHERE Column OPERATOR (SELECT Column FROM table WHERE
Condition)
```

Ex:

Find students who belong to the same department as Alice.

```
SELECT student Name,
FROM Student
WHERE Dept Dept D: C
SELECT Dept ID
FROM Students
WHERE student: 'Alice'
```

);

Answer Output.

Alice

Charlie

old verified

Explanation,

- Inner query: gets Alice's Dept-ID (Say 101).
- Outer query: finds all students with DeptID: 101.
- Correlated Nested Query.
 - The inner query is executed repeatedly, once for each row of the outer query.
 - Inner query depends on Value from the Outer query.

Syntax:

```
SELECT column-list  
FROM table  
WHERE EXISTS  
  SELECT  
    FROM table t2  
  WHERE Exits so  
SELECT 1  
FROM Enrolment  
WHERE Condition involving t1, Column = t2 Column,  
);
```

Examp 2: Correlated Nested Query.

Find student who have enrolled in at least one course

```
SELECT s, Student Name
```

```
FROM Student Name
```

```
WHERE Exist
```

```
SELECT (
```

```
FROM Enrolment s
```

```
WHERE Student ID = s, Student ID
```

```
);
```

Explanation:

- for each student in Outer query the inner checks if that student exists in Enrolments.
- If yes → the student returned.

Answer Output :

Alice

Bob

Charlie

David

VELTECH	
EX No.	
PERFORMANCE (5)	4
RESULT AND ANALYSIS (3)	5
VIVA VOCE (3)	5
RECORD (4)	5
TOTAL (15)	19
SIGN WITH DATE	

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Result :

A list of employees whose salary is greater than the average salary.